

Senex: Unlocking data insights with BigQuery for a more sustainable future with cleaner energy

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)



About Senex

Senex migrates to Cloud for data and learning capabilities to optimize gas well production and its real-time insights. BigQuery, unifying operational and financial data.

Listed on ASX in 1984, Senex is an Australian natural gas producer that supplies the equivalent of 10% of Queensland's domestic gas demand. Keeping a sustainable energy future in mind, the company is focused on delivering low-cost, low-carbon, high-return and long-life natural gas assets. At the 2019 Premier's Awards, the company received a commendation for its innovation in productivity improvement.

Industries: Energy, Chemicals & Utilities

Google Cloud results:

- Improves digital capabilities for cost savings and production efficiencies through current and future machine learning models
- Boosts geospatial data processing by extracting location information from images such as maps and handwritten notes with Cloud Vision AI
- Delivers a single source of truth across the business by interconnecting data sources such as supervisory

Predict
failure
faster
on B

Location: Australia

Storage (<https://cloud.google.com/storage/>),

Servian

About Servian

As an award-winning Google Cloud partner, Servian enables businesses to take advantage of digital transformation. The company offers a wide range of Google Cloud solutions, including digital transformation, marketing analytics, DevOps, hybrid cloud infrastructure, data platforms, artificial intelligence, and machine learning. Servian won the Google Cloud Services Partner of the Year 2019 award for Asia Pacific in recognition of excellent customer service. It employs more than 800 highly skilled technologists in Australia, New Zealand, India, and Singapore.

Google Cloud solutions, including digital transformation, marketing analytics,

Google Cloud (<https://cloud.google.com/>)

BigQuery, and Cloud Infrastructure, data

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Storage (<https://cloud.google.com/storage/>)

Cloud Services Partner of the Year 2019

Document AI

(<https://cloud.google.com/document-ai>)

more than 800 highly skilled technologists

control and
data
acquisition
(SCADA) and
gas sales for
real-time
insights

in Australia, New Zealand, India, and
Singapore.

“Like a tech
startup, our
natural gas
business needs
to be nimble to
capture new
opportunities in
a volatile
market. On
Google Cloud,
we can grow our
business at
scale and
achieve desired
results without
increasing our
cost base.”

*—Timothy Cochrane,
General Manager for
Digital, Senex*

Nations and businesses are increasingly moving from coal to cleaner energy such as liquified natural gas (LNG) to address global warming. Senex (<https://www.senexenergy.com.au/>) is a natural gas supplier in Australia that's ramping up its production to meet energy demand while reducing greenhouse emissions. By creating a data lake on Google Cloud (<https://cloud.google.com/>) to integrate disparate systems, the company aims to unlock data insights to improve operational efficiencies across its value chain, from exploration to production.

"Like a tech startup, our natural gas business needs to be nimble to capture new opportunities in a volatile market," says Timothy Cochrane, General Manager for Digital at Senex. "On Google Cloud, we can grow our business at scale and achieve desired results without increasing our cost base."

"Google Cloud
demonstrated
impressive built-
in OCR

capability. It took
minimal effort
for the team to
pull trapped data
from
handwritten
notes and PDF
documents and
turn it into useful
information to
run our
business. The
difference
between data
and information
is that you can
make business
decisions based
on information.”

*—Timothy Cochrane,
General Manager for
Digital, Senex*

Unlocking data as a business asset

From well-drilling data to transaction data from its trading desk, Senex generates vast amounts of

information each day. According to Cochrane, construction, production, and well data have geospatial components such as maps and latitude and longitude coordinates. For its cloud technology and integration partner selection process, his team set up a proof-of-concept to evaluate machine learning and optical character recognition (OCR) capabilities to maximize the value of data.

“Google Cloud demonstrated impressive built-in OCR capability. It took minimal effort for the team to pull trapped data from handwritten notes and PDF documents and turn it into useful information to run our business,” says Cochrane. “The difference between data and information is that you can make business decisions based on information.”

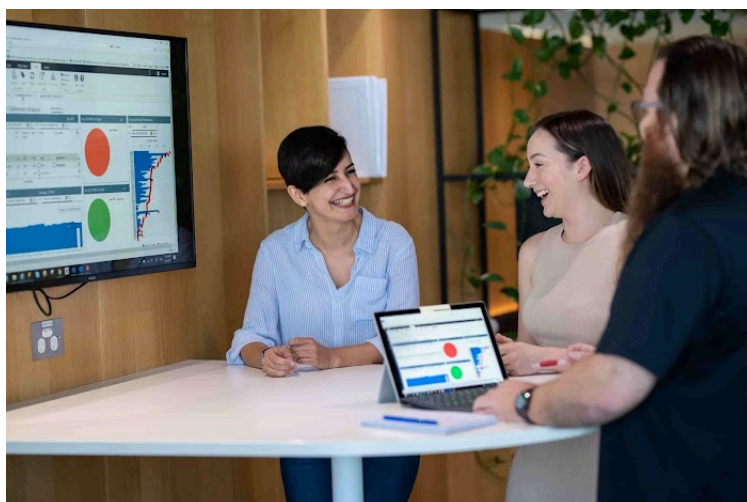
Using Document AI (/document-ai), Senex unlocked data that can be used to optimize well design and operation. As a result, Senex can also look for trends, such as safety incidents across their assets, before determining if the issue needs to be addressed.

“We also liked the element of low code and no-code that Google Cloud presents. As someone from a non-technical background, my concern is business adoption. Advanced analytics can be a powerful tool only if people use it,” says Cochrane.

Having a solid partner is also a critical success factor for digital transformation. Google Cloud partner Servian (<https://www.servian.com/>), played an instrumental role in helping Senex refine its cloud strategy. Instead of a one-size-meets-all approach,

Servian took time to understand how Senex and its people operate and designed a cloud architecture to support its business model. The Servian team then scheduled full-day training workshops and executive briefings to align different stakeholders across their digital transformation journey. To date, up to seven scheduled training days were delivered to 35 staff to ensure that staff were efficiently upskilled and capable of running things on their own.

“As a gas producer, we don’t have an in-house team of data engineers and architects to build a data lake from scratch,” says Cochrane. “Servian provided us with the local resources and analytics expertise to deliver value from digital technologies on Google Cloud in six short months.”



Optimizing production level with predictive maintenance using BigQuery

Senex currently operates over 100 wells and plans to triple its annual gas production by the end of 2025

to meet energy demand. Unplanned downtime caused by pump failures and safety incidents can be costly. At the same time, inefficient pumps consume more energy and are less reliable, both of which result in higher emissions.

However, it's not sustainable for the operations team to visually inspect every well daily and predict what equipment needs to be repaired or replaced. Cochrane's challenge was to simplify complex problems of scale so that two operators and a production engineer could continuously manage a growing number of wells.

Instead of waiting for an asset to fail before fixing it, Senex deploys a machine learning model to identify problems before they occur and apply remedial action to prevent pump failure. A data pipeline ingests live data from SCADA Historian application data into BigQuery (/bigquery) tables, which becomes the source for the TensorFlow machine learning model to predict torque events. Prior to Google Cloud, Senex was running on historical data only. Moving to the cloud allowed the business to run their model in near real time which helps identify issues before they lead to operational issues. Previously, Senex could only query a limited amount of production data at any given time on a dedicated server. Analyzing data from wells was previously done manually. With the help of Google Cloud, it now runs every 15 minutes across its Atlas field, with plans to expand to the remaining assets.

"With its ability to query data at infinite scale, BigQuery is perfect for analyzing massive datasets.

It chews data for breakfast and spits out insights in seconds,” says Cochrane. Senex now handles 2.67 TB of data daily, and speeds up analysis to under four minutes. Every morning, operators can look at a map of the wells, identify the high-risk ones in red, and prioritize urgent work for the day.

“Leveraging
Google Cloud
technology
allows us to
work smarter.
We can uncover
insights from
interconnected
datasets to
make confident
decisions to
drive our
business
forward for a
sustainable
future.”

—*Timothy Cochrane,*
General Manager for
Digital, Senex

Reliable forecasting on data lake for future business growth

In the near term, Senex wants to support its production team to maximize the production from each well. In the long run, the company intends to leverage the power of Google Cloud to drive a data-driven culture with advanced analytics.

Capital costs such as construction and procurement make up a large chunk of the oil and gas business. It is crucial for Senex to know where it can save energy, time, and cost to pursue growth projects. Digital projects focused on enabling Senex to work smarter are expected to realize financial benefits up to \$60 million over the next five years.

Cochrane adds, “Leveraging Google Cloud technology allows us to work smarter. We can uncover insights from interconnected datasets to make confident decisions to drive our business forward for a sustainable future.”